Direct CO_2 emissions from fuel combustion by country GRI 305-1

Million metric tons	2019	2018	2017
Russia	24.9	25.3	26.4
Germany	11.1	17.2	16.9
United Kingdom	5.6	7.6	6.2
Netherlands	3.2	5.5	7.9
France	1.0	2.9	5.1
Hungary	0.9	0.8	0.8
Czech Republic[1]	0.1	<0.1	-
Sweden	<0.01	<0.01	<0.01
Total	**47.0**	**59.5**	**63.3**

Operational control approach taken – figures (100% of direct emissions of the facility) from any generation assets over which Uniper has operational control. Data for all countries except Russia were determined using the European Union Emissions Trading Scheme rules. Rounding corrections per country result in small differences which are considered in the total sum. The carbon emissions for France are for the period January 1 to June 30, 2019 only.

[1] Emissions in the Czech Republic were inadvertently excluded from the operational control approach in 2018. These have now been added for 2018 and 2019.

Under our ambitious timeline, we'll close about 1.5 GW of coal-fired capacity in Germany by year-end 2022 and intend to submit another 1.4 GW to the federal government's shutdown scheme by 2025. We had already withdrawn just over 2.4 GW through year-end 2015. Together, these closures will yield total carbon savings of up to 18 million metric tons per year.

Datteln 4, our new hard-coal-fired power plant in west-central Germany, is scheduled to enter service in the summer of 2020. There has been some criticism regarding the commissioning of Datteln 4 along with the decision on a coal exit in Germany. However, Datteln 4 is significantly more efficient than older coal-fired power plants. When cogenerating heat for 100,000 households in the region it will have a fuel efficiency of nearly 60%, similar to that of a gas turbine. Along with the aforementioned plant closures, having Datteln 4 online will enable us to reduce our carbon emissions in Germany by 40% by the end of 2025. This is why, after careful consideration, the German federal government decided that the country could decarbonize more quickly by having Datteln 4 become operational and displace the production of older, less efficient coal-fired plants.

In February 2020, we signed an agreement to sell our 58% stake in Schkopau, a lignite-fired power plant in Saxony-Anhalt in eastern Germany, to Saale Energie GmbH, a subsidiary of Czech energy producer EPH, which owns the other 42%. The transfer of ownership will take place in October 2021. It will mark the end of our lignite-fired power generation in Europe.

Coal is still part, but now a steadily shrinking part, of our portfolio. Roughly two-thirds of our total electricity and heat output already comes from low-emission hydro, nuclear, and gas. As we close coal-fired plants, this proportion will increase and our carbon intensity will decrease, bringing us progressively closer to our goal of carbon neutrality.

Gas, the cleanest fossil fuel, has a unique ability to deliver deep and rapid emissions reductions across many sectors. Gas is the fastest way to decarbonize the power industry. Shifting from coal and petroleum products to gas will rapidly reduce emissions in heavy industry, space heating, and transport. Multi-sector conversion to gas is the fastest and cheapest way for Europe to reduce its carbon emissions by up to 65% in the next two decades. Gas will also serve as a vital emissions-reducing technology today while transformative innovations, like synthetic fuels, are developed and scaled up.

Natural gas can play its role in the energy transition even more effectively if it becomes more climate friendly, for example, when equipment to produce hydrogen and methane is powered by renewable electricity in a process known as power-to-gas. Alternative fuels can help make mobility more sustainable as well. Uniper is therefore developing businesses for several alternative fuels, including liquefied natural gas (LNG) as a fuel for heavy trucks. LNG trucks have lower emissions and are quieter.

Greenhouse Gas Protocol Scope 2 and 3

GRI 305-2/3 Our Scope 2 indirect emissions totaled 1.12 million metric tons of CO_2 (2018: 1.10 million metric tons of CO_2) and 1.57 million metric tons CO_2 (2018: 1.67 million metric tons of CO_2) using the location-based method and market-based method, respectively. Our Scope 2 emissions now include indirect emissions from purchased electricity used for pump storage in our hydro plants in Germany. The 2018 figures were also updated. Our Scope 3 indirect emissions related to extraction and transportation of consumed fuels totaled 8.7 million metric tons of CO_2, lower than in 2018 (10.3 million metric tons).

A higher CDP score

In 2019, Uniper responded to CDP's sector-specific climate change questionnaire, about Uniper's performance in the 2018 calendar year. CDP, formerly known as the Carbon Disclosure Project, runs a global disclosure system for companies to improve awareness through measurement and disclosure of environmental data, climate risks, and carbon management. After evaluating our responses, CDP gave Uniper a score of B- which is an improvement on the previous year's score (C). Scores range from A (best) to F. CDP raised the score because it recognized that we had identified and realized potential for improvement in our sustainability performance. We will continue our efforts in 2020 and are evaluating actions that could further improve our CDP score.

Climate Action & Strategy Team

GRI 103-1 and GRI 305-5 Uniper aims to be a pacesetter in decarbonizing the energy it supplies. To help us play this role as effectively as possible, in December 2019 we formed a panel of experts, the Climate Action & Strategy Team. Led by the HSSE & Sustainability and Corporate Strategy departments, it brings together representatives of other departments that are integral to setting our decarbonization course.

The team's first task is to carefully analyze Uniper's current emission baseline by geography, fuel type, and emission scope (1, 2, and 3). The next – as part of the implementation of our new strategy – is to identify the various decarbonization options available to us in power generation and commodity trading and assess their potential impact on our carbon footprint, operations, and bottom line. Bringing these two sets of information together – detailed emissions data and a list of decarbonization options – will enable the Uniper Management Board to set specific ambitious emission reduction targets per business that are economically viable and enable us to continue to provide a reliable supply of energy to customers.

The team's third task is to define Uniper's future approach to emission disclosures. Transparency is integral to our climate strategy and will guide this approach. As stated above, Uniper has already improved its CDP score, and the team will explore ways to further improve this score and transparency generally. As part of this effort, the team will continue to assess whether, and to what degree, the framework for voluntary climate-related financial risk disclosures issued by the Task Force on Climate-related Financial Disclosures (TCFD) could fit with Uniper's approach to emission disclosures.

Innovations for a low-carbon future

GRI 103-1 Innovation and cutting-edge technologies are crucial to our ability to Empower Energy Evolution. They make our existing businesses more efficient, competitive, and sustainable. Also, they enable us to establish new businesses that we believe could accelerate the transition to a carbon-neutral future. That's why we continually track and analyze emerging technologies. Our innovation strategy reflects the three pivotal trends that are transforming the energy industry: decarbonization, the decentralization of energy generation and supply, and digitalization. We have the assets and energy IQ to shape these trends in a way that creates value for our company and for society.

How we manage innovation

GRI 103-2/3 and 302-2/4 G4-DMA Uniper develops innovative, scalable business models in a variety of areas. Some of them – like a flexible power supply, utility-scale batteries, and other new storage technologies – demonstrably enhance the energy system's ability to add more renewable energy sources while maintaining system stability. Although crucial to the success of the energy transition, they're incremental in nature: they'll help the energy transition do what it's already doing even better. Other technologies like green hydrogen and carbon recycling have the potential to be truly game-changing. Because they can do something special: go a long way towards making emission-heavy but hard-to-decarbonize industries – particularly chemicals, and air, maritime, and heavy-road transport – much cleaner.

Uniper has invested in a number of pilot projects to refine, scale up, and deploy a variety of technologies on a commercial scale. In addition, we've set a target of conducting, by 2022, at least 20 projects, the main aim of which include decarbonization. At year-end 2019, Uniper was working on 12 such projects.

Employees at Uniper's power-to-gas plant in Falkenhagen, Germany.

The European Union and many of its member states have committed to becoming as climate neutral as possible by 2050. This will require a lot more renewables. But it will also require many of the technologies mentioned, from utility-scale batteries to green hydrogen, in which Uniper is a pacesetter. Our innovation portfolio is focused on issues where we can best leverage our existing capabilities and assets to accelerate the transition to a low-carbon – and, ultimately – a carbon-neutral future.

New flexibility

Flexibility supports the transition to a low-carbon energy world in two ways. First, it balances out the fluctuations in renewables output; this capability will help support the integration of large amounts of renewables capacity. Second, the flexibility provided by energy storage or conversion is able to capture more of this output. We're pioneering the development of innovative technologies for both forms of flexibility.

Converting green energy into green gas

On particularly windy, sunny days, wind and solar farms sometimes have to curtail and even suspend production because of grid congestion. That's green energy gone to waste. Power-to-gas (P2G) can help: instead of taking wind turbines or solar arrays offline when the grid is congested, their output can be used to run on-site electrolysis equipment that produces green hydrogen, which is then injected into the gas pipeline system.

Key Figures GRI 102-8, 303-3, 305-1/2/3/4/7

Indicators	unit	2019	2018
Uniper employees[1]		11,532	11,780
Proportion of female employees	%	24.6	24.2
Combined TRIF[2]		1.48	1.47
Uniper generating capacity[3]	GW	34.3	36.6
Average asset availability of our conventional generation fleet	%	79.1	79.0
Unplanned unavailability of our conventional generation fleet	%	12.0	11.6
Coal consumption[4]	m metric tons	11.8	16.7
Gas volume sold	billion kWh	2,179.3	2,019.3
Direct scope 1 emissions[5]	m metric tons of CO_2	47.0	59.5
Indirect scope 2 emissions (location-based method)[5]	m metric tons of CO_2	1.12	1.10
Indirect scope 2 emissions (market-based method)[5]	m metric tons of CO_2	1.57	1.67
Average carbon intensity (threshold commitment period 2018-2020)[6]	g/kWh	474	499
PFA and FBA produced	m metric tons	0.93	1.44
Gypsum produced[7]	m metric tons	0.6	0.9
Facilities certified to ISO 14001[5]	%	100	100
Facilities certified to OHSAS18001[5]	%	100	100
Cooling water withdrawal	bn m^3	4.0	4.3
SO_2 emissions	kt	12.0	18.4
NOx emissions	kt	47.3	57.4
Dust emissions	kt	1.53	1.57
Severe environmental incidents[8]		0	0

[1] Headcount as of December 31, 2019. Figures do not include board members, managing directors, apprentices, work-study students and interns.
[2] Total recordable incidents per million hours of work (combined TRIF) for Uniper Group employees and contractors engaged by Uniper. Combined TRIF takes account of all relevant reports, including those from Uniper companies that are not fully consolidated but in which Uniper SE has operational control.
[3] Net capacity as of December 31, 2019 (accounting view).
[4] Figure includes domestic lignite consumed by Unipro plants. 2018 Russian figures corrected. 2019 figures from France calculated using DEFRA emission factors.
[5] These figures encompass all consolidated Uniper entities as well as nonconsolidated entities over which we have operational control.
[6] We calculate carbon intensity using the financial control approach. This means that our carbon intensity is the ratio between the direct CO_2 emissions from our fully consolidated, stationary fossil-fueled power plants and power-and-heat plants, and these plants' power and heat output. It does not include plants that produce heat/steam only.
[7] 2018 figures from German power plants corrected.
[8] Severe impact beyond site which is reversible within years or irreversible.

Direct CO_2 emissions from fuel combustion by country GRI 305-1

Million metric tons	2019	2018	2017
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Operational control approach taken - figures (100% of direct emissions of the facility) from any generation assets over which Uniper has operational control. Data for all countries except Russia were determined using the European Union Emissions Trading Scheme rules. Rounding corrections per country result in small differences which are considered in the total sum.

[1] Emissions in the Czech Republic were inadvertently excluded from the operational control approach in 2018. These have now been added for 2018 and 2019.

Indirect CO_2 emissions[1] GRI 305-2

Greenhouse Gas Protocol Scope 2

	metric t CO_2	2019	2018
Location-based method	Indirect emissions from purchased electricity	1,023,106	1,008,531
	Indirect emissions from heat and cooling	106,171	92,501
	Total	**1,129,277**	**1,101,032**
Market-based method	Indirect emissions from purchased electricity	1,467,501	1,584,188
	Indirect emissions from heat and cooling	106,171	92,501
	Total	**1,573,672**	**1,676,689**

[1] These figures include emissions from consolidated and non-consolidated generation assets over which Uniper has operational control. Indirect emissions from purchased electricity used for pump storage in our hydro plants in Germany has been included for 2019 and updated for the 2018 figures. Figures for electricity consumption from Uniper's Düsseldorf offices were not available for 2019 and therefore are not included.

Indirect CO_2 emissions[1] GRI 305-3

Greenhouse Gas Protocol Scope 3

m metric t CO_2	2019	2018
Upstream indirect Scope 3 CO_2 emissions	8.7	10.3

[1] Calculation of upstream Scope 3 emissions associated with extraction, refining and transportation of the raw fuel sources to an organization's site (or asset), prior to combustion using well-to-tank (DEFRA) fuel conversion factors.

Fully consolidated generating capacity by technology[1]

MW	2019	2018	2017
Gas	17,439	18,916	18,917
Coal	9,135	10,345	10,325
Hydro	3,570	3,570	3,567
Nuclear	1,400	1,400	1,400
Other	2,801	2,358	2,229
Total	**34,345**	**36,589**	**36,438**

[1] Net capacity as of December 31, 2019 (accounting view).

Power production

By primary energy source

Billion kWh	2019	2018	2017
Gas[1]	60.3	60.5	61.9
Coal	19.9	31.8	35.8
Nuclear	11.0	10.7	11.1
Hydro	12.7	10.3	11.8
Other renewables[2]	0.1	0.2	0.2
Biomass	0.0	0.3	0.0
Total[3]	**104.0**	**113.9**	**120.8**

[1] Figures include production from oil.

[2] Figures include production from non-material wind and solar assets (aggregated installed capacity 95 MW).

[3] Possible rounding differences between individual figures and totals.

Average asset availability[1]

%	2019	2018
Average asset availability in Europe and Russia	79.1	79.0

[1] Availability is equal to 100% minus planned and unplanned unavailability. Uniper Group figures are volume-based weighted averages. They refer to Uniper's current operational portfolio and is based on the proportion of our stake in an asset (2018 figures adjusted). Assets in France are included from January 1 to June 30, 2019.

Health and safety GRI 403-1

Total recordable incident frequency (TRIF)

	2019	2018
Combined TRIF[1]	1.48	1.47
Employee TRIF	0.98	0.90
Contractor TRIF	2.05	2.18
Combined LTIF	1.05	0.96
Employee LTIF	0.93	0.57
Contractor LTIF	1.19	1.44

[1] Total recordable incidents per million hours of work (TRIF) for Uniper employees and contractors engaged by Uniper. TRIF takes account of all relevant reports, including those from Uniper companies that are not fully consolidated but in which Uniper SE has operational control.

New hires from external market GRI 401-1

By age range and gender

	Male				Female				Total	
Employee structure	**2019**		**2018**		**2019**		**2018**		**2019**	**2018**
Age range	Amount	Share	Amount	Share	Amount	Share	Amount	Share	Amount	Amount
< 21	89	80.9	98	80.3	21	19.1	24	19.7	110	122
21 – 30	336	68.3	307	65.9	156	31.7	159	34.1	492	466
31 – 40	262	69.3	142	61.2	116	30.7	90	38.8	378	232
41 – 50	136	64.8	107	68.6	74	35.2	49	31.4	210	156
51 – 60	92	65.7	57	68.7	48	34.3	26	31.3	140	83
> 60	36	78.3	18	81.8	10	21.7	4	18.2	46	22
Total	**951**	**69.1**	**729**	**67.4**	**425**	**30.9**	**352**	**32.6**	**1,376**	**1,081**

[1] Permanents + temporary staff + managing directors/board members + interns/working students + apprentices.